Staggered mesh method for MP3 perturbation energy calculation

Xin Xing

September 15, 2022

1 Introduction

2 Theory

Consider a generic periodic system. Let Ω be its unit cell, Ω^* be the associated first Brillouin zone (BZ), $\mathbb{L}$ be the Bravais lattice, and $\mathbb{L}^*$ be the associated reciprocal lattice. First sample a Monkhorst-Pack (MP) mesh $\mathcal{K}$ with $N_{\mathbf{k}}$ points in Ω^*. A model Hamiltonian can then be defined on $\mathcal{K}$ and solving it by the Hartree-Fock method computes a set of single-particle molecular orbitals and orbital energies $\{\psi_{n\mathbf{k}}, \varepsilon_{n\mathbf{k}}\}$ indexed by band index n and momentum vector $\mathbf{k} \in \mathcal{K}$. In this paper, we always assume all the orbitals and orbital energies to be exact, i.e., converged in the TDL.

For brevity, we use capital letters to denote the band index and momentum vector pair, e.g., $P \leftrightarrow (p, \mathbf{k}_p)$, and use $P \in \{I, J, K, L\}$ to refer to occupied orbitals and $P \in \{A, B, C, D\}$ to refer to unoccupied ones. The normalized electron repulsion integral (ERI) in the molecular orbital basis can be computed as

$$\langle PQ|RS\rangle = \frac{1}{|\Omega|} \sum_{\mathbf{G}\in\mathbb{L}^*}{}' \frac{4\pi}{|\mathbf{k}_r - \mathbf{k}_p + \mathbf{G}|^2} \hat{\varrho}_{p\mathbf{k}_p, r\mathbf{k}_r}(\mathbf{G}) \hat{\varrho}_{q\mathbf{k}_q, s\mathbf{k}_s}(\mathbf{G}^{\mathbf{k}_r,\mathbf{k}_s}_{\mathbf{k}_p,\mathbf{k}_q} - \mathbf{G}), \tag{2.1}$$

where $\mathbf{k}_r - \mathbf{k}_q$ is referred to as the momentum transfer vector, $\mathbf{G}^{\mathbf{k}_r,\mathbf{k}_s}_{\mathbf{k}_p,\mathbf{k}_q} = \mathbf{k}_p + \mathbf{k}_q - \mathbf{k}_r - \mathbf{k}_r \in \mathbb{L}^*$ by crystal momentum conservation, $\sum'_{\mathbf{G}\in\mathbb{L}^*}$ excludes possible term with $\mathbf{k}_r - \mathbf{k}_q + \mathbf{G} = \mathbf{0}$, and $\hat{\varrho}_{p\mathbf{k}_p, r\mathbf{k}_r}(\mathbf{G}) = \langle \psi_{p\mathbf{k}_p} | e^{\mathrm{i}(\mathbf{k}_p - \mathbf{k}_r - \mathbf{G})\mathbf{r}} | \psi_{r\mathbf{k}_r} \rangle$ with $\mathbf{G} \in \mathbb{L}^*$ denotes the pair product.

The MP3 perturbation energy with $\mathbf{k}$-point mesh $\mathcal{K}$ is defined as

$$E_{\text{MP3}}(N_{\mathbf{k}}) = \frac{1}{N_{\mathbf{k}}^3} \sum_{IJAB} \left(2\langle IJ|AB\rangle - \langle IJ|BA\rangle\right) t_{IJ}^{AB}(N_{\mathbf{k}}), \tag{2.2}$$

where the summation over $P \in \{I, J, A, B\}$ corresponds to summing p over occupied/virtual band indices and $\mathbf{k}_p$ over all point in $\mathcal{K}$, and a unique $\mathbf{k}_b \in \mathcal{K}$ is determined by the crystal momentum conservation given each triplet $(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a)$.

The normalized MP3 amplitude with $\mathbf{k}$-point mesh $\mathcal{K}$ above is defined as

$$t_{IJ}^{AB}(N_{\mathbf{k}}) = s_{IJ}^{AB} + \frac{1}{\varepsilon_{IJAB}}\left[\frac{1}{N_{\mathbf{k}}}\sum_{KL}\langle KL|IJ\rangle\, s_{KL}^{AB} + \frac{1}{N_{\mathbf{k}}}\sum_{CD}\langle AB|CD\rangle\, s_{IJ}^{CD} + \right. \tag{2.3}$$
$$\left. + \mathcal{P}\left(\frac{1}{N_{\mathbf{k}}}\sum_{KC}(2\langle AK|IC\rangle - \langle AK|CI\rangle)s_{KJ}^{CB} - \langle AK|IC\rangle\, s_{KJ}^{BC} - \langle AK|CJ\rangle\, s_{KI}^{BC}\right)\right],$$

where $\varepsilon_{IJAB} = \varepsilon_I + \varepsilon_J - \varepsilon_A - \varepsilon_B$, $s_{IJ}^{AB} = \frac{1}{\varepsilon_{IJAB}}\langle AB|IJ\rangle$ is the normalized MP2 amplitude and $\mathcal{P}$ is a permutation operator defined as $\mathcal{P}(\cdots)_{IJ}^{AB} = (\cdots)_{IJ}^{AB} + (\cdots)_{JI}^{BA}$.

2.1 Error analysis

In the TDL with $\mathcal{K} \to \Omega^*$, we have $\frac{1}{N_{\mathbf{k}}}\sum_{\mathbf{k}\in\mathcal{K}} \to \frac{1}{|\Omega^*|}\int_{\Omega^*} \mathrm{d}\mathbf{k}$ and the MP3 perturbation energy in Eq. (2.2) converges to

$$E_{\text{MP3}}^{\text{TDL}} = \frac{1}{|\Omega^*|^3}\int_{\Omega^*\times\Omega^*\times\Omega^*} \mathrm{d}\mathbf{k}_i\,\mathrm{d}\mathbf{k}_j\,\mathrm{d}\mathbf{k}_a \sum_{ijab}\left(2\langle IJ|AB\rangle - \langle IJ|BA\rangle\right) t_{IJ}^{AB}(\infty). \tag{2.4}$$

The MP3 amplitude in the TDL, $t_{IJ}^{AB}(\infty)$, can be similarly formulated based on Eq. (2.3) as

$$t_{IJ}^{AB}(\infty) = s_{IJ}^{AB} + \frac{1}{\varepsilon_{IJAB}}\left[\frac{1}{|\Omega^*|}\int_{\Omega^*} \mathrm{d}\mathbf{k}_k \sum_{kl}\langle KL|IJ\rangle\, s_{KL}^{AB} + \frac{1}{|\Omega^*|}\int_{\Omega^*} \mathrm{d}\mathbf{k}_c \sum_{cd}\langle AB|CD\rangle\, s_{IJ}^{CD} + \right.$$
$$\left. + \mathcal{P}\left(\frac{1}{|\Omega^*|}\int_{\Omega^*} \mathrm{d}\mathbf{k}_k \sum_{kc}(2\langle AK|IC\rangle - \langle AK|CI\rangle)s_{KJ}^{CB} - \langle AK|IC\rangle\, s_{KJ}^{BC} - \langle AK|CJ\rangle\, s_{KI}^{BC}\right)\right].$$

By comparing Eq. (2.2) and Eq. (2.4), the finite size error in MP3 calculation can then be split as

$$E_{\text{MP3}}^{\text{TDL}} - E_{\text{MP3}}(N_{\mathbf{k}}) = \left(\frac{1}{|\Omega^*|^3}\int_{(\Omega^*)^{\times 3}} \mathrm{d}\mathbf{k}_i\,\mathrm{d}\mathbf{k}_j\,\mathrm{d}\mathbf{k}_a - \frac{1}{N_{\mathbf{k}}^3}\sum_{\mathbf{k}_i\mathbf{k}_j\mathbf{k}_a\in\mathcal{K}}\right)\sum_{ijab}\left(2\langle IJ|AB\rangle - \langle IJ|BA\rangle\right) t_{IJ}^{AB}(\infty)$$
$$+ \frac{1}{N_{\mathbf{k}}^3}\sum_{\mathbf{k}_i,\mathbf{k}_j,\mathbf{k}_a\in\mathcal{K}}\left(2\langle IJ|AB\rangle - \langle IJ|BA\rangle\right)\left(t_{IJ}^{AB}(\infty) - t_{IJ}^{AB}(N_{\mathbf{k}})\right)$$

which is made up by two separate parts: the quadrature error in evaluating the energy with exact amplitudes and the error in evaluating each amplitude.

Following a similar analysis as our earlier work [1], we can show that the finite size error in MP3 energy calculation with exact amplitudes scales as $\mathcal{O}(N_{\mathbf{k}}^{-1})$ and comes from two sources: the omission of singular terms in ERIs with zero momentum transfer and the quadrature error in volume elements near zero moment transfer. For example, $\langle IJ|AB\rangle$ in Eq. (2.4) as a function of $\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a$ contains a singular term

$$\frac{4\pi}{|\mathbf{q}|^2}\hat{\varrho}_{i\mathbf{k}_i,a(\mathbf{k}_i+\mathbf{q})}(\mathbf{0})\hat{\varrho}_{j\mathbf{k}_j,b(\mathbf{k}_j-\mathbf{q})}(\mathbf{0}) \sim \frac{\mathcal{O}(\mathbf{q}^2)}{|\mathbf{q}|^2}, \quad \mathbf{q} = \mathbf{k}_a - \mathbf{k}_i,$$

that is discontinuous at $\mathbf{q} = \mathbf{0}$. This singular term at $\mathbf{q} = \mathbf{0}$ is directly neglected in the energy calculation, equivalent to neglecting the integral of this term in the $\frac{|\Omega^*|}{N_{\mathbf{k}}}$-volume element containing $\mathbf{q} = \mathbf{0}$. The neglected

integral here is of scale $\mathcal{O}(N_{\mathbf{k}}^{-1})$ and contributes to the quadrature error. More delicate analysis can further show that the quadrature error in volume elements near $\mathbf{q} = \mathbf{0}$ generally scales as $\mathcal{O}(N_{\mathbf{k}}^{-1})$ but in cases with certain high symmetries can be $o(N_{\mathbf{k}}^{-1})$.

The finite size error in evaluating MP3 amplitude can be analyzed similarly. For example, the 4h2p term in Eq. (2.3) converges in the TDL as

$$\frac{1}{N_{\mathbf{k}}} \sum_{KL} \langle KL|IJ\rangle \, s_{KL}^{AB} \xrightarrow{N_{\mathbf{k}}\to\infty} \frac{1}{|\Omega^*|} \int_{\Omega^*} \mathrm{d}\mathbf{k}_k \sum_{kl} \langle KL|IJ\rangle \, s_{KL}^{AB}. \tag{2.5}$$

Its finite size error can also be interpreted as quadrature error and scales as $\mathcal{O}(N_{\mathbf{k}}^{-\frac{1}{3}})$ due to the $1/|\mathbf{q}|^2$-singularity near zero momentum transfer of involved ERIs $\langle KL|IJ\rangle$ with $k = i$ and $l = j$. Similarly, there is also $1/|\mathbf{q}|^2$ singularity and $\mathcal{O}(N_{\mathbf{k}}^{-\frac{1}{3}})$ quadrature error in another five terms:

$$\frac{1}{N_{\mathbf{k}}} \sum_{CD} \langle AB|CD\rangle \, s_{IJ}^{CD}, \quad \mathcal{P}\frac{1}{N_{\mathbf{k}}} \sum_{KC} \langle AK|CI\rangle \, s_{KJ}^{CB}, \quad \mathcal{P}\frac{1}{N_{\mathbf{k}}} \sum_{KC} \langle AK|CJ\rangle \, s_{KI}^{BC} \tag{2.6}$$

when the corresponding band indices match, e.g., $a = c, b = d$ for the first term. All the other terms in the amplitude have singularities of form $\frac{\mathcal{O}(\mathbf{q}^2)}{|\mathbf{q}|^2}$ near zero momentum transfer and thus have $\mathcal{O}(N_{\mathbf{k}}^{-1})$ quadrature error. Overall, the finite size error in each amplitude term $t_{IJ}^{AB}(N_{\mathbf{k}})$ scales as

$$|t_{IJ}^{AB}(N_{\mathbf{k}}) - t_{IJ}^{AB}(\infty)| = \mathcal{O}(N_{\mathbf{k}}^{-\frac{1}{3}}) \qquad \forall I, J, A, B.$$

2.2 Madelung constant correction

In this section, we introduce the Madelung constant correction to remove the leading $\mathcal{O}(N_{\mathbf{k}}^{-\frac{1}{3}})$ quadrature error in MP3 amplitude calculation. The Madelung constant correction is commonly used to reduce the finite size error in exchange energy calculation by

$$E_X(N_{\mathbf{k}}) = -\frac{1}{N_{\mathbf{k}}^2} \sum_{IJ} \langle IJ|JI\rangle \longrightarrow E_X^{\text{corrected}}(N_{\mathbf{k}}) = -\frac{1}{N_{\mathbf{k}}^2} \sum_{IJ} \langle IJ|JI\rangle + N_{\text{occ}}\xi,$$

where the Madelung constant ξ is formally defined as

$$\xi = \left(\frac{1}{N_{\mathbf{k}}} \sum_{\mathbf{q}\in\mathcal{K}_{\mathbf{q}}} -\frac{1}{|\Omega^*|} \int_{\Omega^*} \mathrm{d}\mathbf{q} \right) \sum_{\mathbf{G}\in\mathbb{L}^*} \frac{4\pi}{|\Omega|} \frac{1}{|\mathbf{q}+\mathbf{G}|^2}. \tag{2.7}$$

A key observation is that the integrand in E_X, $\sum_{ij} \langle IJ|JI\rangle$, has leading singular term

$$N_{\text{occ}} \frac{4\pi}{|\Omega|} \frac{1}{|\mathbf{q}|^2}, \quad \mathbf{q} = \mathbf{k}_j - \mathbf{k}_i,$$

near $\mathbf{q} = \mathbf{0}$ which exactly matches N_{occ} times the singular part of the function in ξ. From a numerical quadrature perspective, the Madelung constant correction is equivalent to a singularity subtraction method which subtracts the quadrature of $N_{\text{occ}} \frac{4\pi}{|\Omega|} \frac{1}{|\mathbf{q}|^2}$ from the calculation and replaces it by its exact integral. After this correction, the dominate quadrature error comes from the second-leading singular terms in the integrand

that is of form $\mathcal{O}(\mathbf{q}^2)/|\mathbf{q}|^2$. This remaining error can be proved to scale as $\mathcal{O}(N_{\mathbf{k}}^{-1})$. For more details of the above analysis, readers can refer to our earlier work [1].

Following the same idea, we can use the Madelung constant correction to remove the $\mathcal{O}(N_{\mathbf{k}}^{-\frac{1}{3}})$ quadratuer error in amplitude calculation due to all the $\mathcal{O}(1)/|\mathbf{q}|^2$ singular terms involved. We note that each of the amplitude terms in Eq. (2.5) and Eq. (2.6) has its leading singular term as $s_{IJ}^{AB}\frac{4\pi}{|\Omega|}\frac{1}{|\mathbf{q}|^2}$. Thus, we can apply the Madelung constant correction $s_{IJ}^{AB}\xi$ separately to each of the six terms as

$$\begin{aligned}
\frac{1}{N_{\mathbf{k}}}\sum_{KL}\langle KL|IJ\rangle\, s_{KL}^{AB} &\longrightarrow \frac{1}{N_{\mathbf{k}}}\sum_{KL}\langle KL|IJ\rangle\, s_{KL}^{AB} - s_{IJ}^{AB}\xi,\\
\frac{1}{N_{\mathbf{k}}}\sum_{CD}\langle AB|CD\rangle\, s_{IJ}^{CD} &\longrightarrow \frac{1}{N_{\mathbf{k}}}\sum_{CD}\langle AB|CD\rangle\, s_{IJ}^{CD} - s_{IJ}^{AB}\xi,\\
\frac{1}{N_{\mathbf{k}}}\sum_{KC}\langle AK|CI\rangle\, s_{KJ}^{CB} &\longrightarrow \frac{1}{N_{\mathbf{k}}}\sum_{KC}\langle AK|CI\rangle\, s_{KJ}^{CB} - s_{IJ}^{AB}\xi,\\
\frac{1}{N_{\mathbf{k}}}\sum_{KC}\langle AK|CJ\rangle\, s_{KI}^{BC} &\longrightarrow \frac{1}{N_{\mathbf{k}}}\sum_{KC}\langle AK|CJ\rangle\, s_{KI}^{BC} - s_{IJ}^{AB}\xi,
\end{aligned}$$

and another two terms and their corrections are from applying $\mathcal{P}$ to the last two terms above. These corrections also correspond to adding a Madelung constant shift to the Ewald kernel in ERI computation. Overall, the Madelung constant correction to the MP3 amplitude calculation writes as

$$\widetilde{t}_{IJ}^{AB}(N_{\mathbf{k}}) = t_{IJ}^{AB}(N_{\mathbf{k}}) + \frac{2}{\varepsilon_{IJAB}}s_{IJ}^{AB}\xi.$$

Mathematically, we could prove that the Madelung constant correction can reduce the quadrature error in MP3 amplitude from $\mathcal{O}(N_{\mathbf{k}}^{-\frac{1}{3}})$ to $\mathcal{O}(N_{\mathbf{k}}^{-1})$, i.e.,

$$|\widetilde{t}_{IJ}^{AB}(N_{\mathbf{k}}) - t_{IJ}^{AB}(\infty)| = \mathcal{O}(N_{\mathbf{k}}^{-1}) \qquad \forall I, J, A, B.$$

After the correction, the leading $\mathcal{O}(N_{\mathbf{k}}^{-1})$ quadrature error in $\widetilde{t}_{IJ}^{AB}(N_{\mathbf{k}})$ comes from two similar sources as in the energy calculation: the omission of singular terms in ERIs with zero momentum transfer and the quadrature error in volume elements near zero moment transfer. Detailed justification of the above Madelung constant correction to MP3 amplitude calculation will be presented in a later work.

3 Staggered mesh method

From the above analysis, one main source of finite size error in MP3 calculation is the omission of singular terms when evaluating ERIs with zero momentum transfer, which leads to $\mathcal{O}(N_{\mathbf{k}}^{-1})$ error both in energy calculation with exact amplitude and in amplitude calculation with the Madelung constant correction. This is because standard MP3 calculation $E_{\text{MP3}}(N_{\mathbf{k}})$ has all momentum vectors sampled on the same mesh $\mathcal{K}$ and always triggers ERIs with zero momentum transfer, e.g., $\langle IJ|AB\rangle$ with $\mathbf{k}_i = \mathbf{k}_a \in \mathcal{K}$. From a numerical quadrature perspective, $E_{\text{MP3}}(N_{\mathbf{k}})$ is only one trapezoidal quadrature rule that approximates $E_{\text{MP3}}^{\text{TDL}}$ as an integral. In the most general case, we can choose arbitrary uniform meshes for each of the momentum vectors in the definition of $E_{\text{MP3}}(N_{\mathbf{k}})$ and define a different trapezoidal quadrature rule that also converge to the same integral.

Based on these observations, to avoid the finite size error due to zero momentum transfer, all we have to do is to sample momentum vectors in $E_{\text{MP3}}(N_{\mathbf{k}})$ on different uniform meshes that can avoid ERIs with zero momentum transfer. This idea, referred to as the staggered mesh method, has been applied in our early works on MP2 and RPA energy calculations where two uniform meshes are used for occupied and virtual orbitals, respectively. In this way, all the ERIs with zero momentum transfer included in MP2 and RPA calculations (that are of form $\langle oo|vv\rangle$) are avoided, and the finite size errors can be significantly reduced. In this section, we generalize the staggered mesh method to MP3 calculation.

First thing to note is that using two staggered meshes is no longer sufficient to help avoid all ERIs with zero momentum transfer in MP3 calculation. This can be easily justified by checking the MP3 4h2p direct energy term:

$$\frac{1}{N_{\mathbf{k}}^3}\sum_{IJABKL} 2\langle IJ|AB\rangle\langle KL|IJ\rangle\langle AB|KL\rangle\frac{1}{\varepsilon_{IJAB}\varepsilon_{KLAB}}.$$

where the three involved ERIs requires I, A, K to be sampled on three different meshes to avoid zero momentum transfer. As to be demonstrated next, three staggered uniform meshes are sufficient for MP3 energy calculation.

Given an arbitrary uniform mesh $\mathcal{K}$, we construct three staggered meshes $\mathcal{K}_0, \mathcal{K}_{\frac{1}{3}}, \mathcal{K}_{\frac{2}{3}}$ by shifting $\mathcal{K}$ by 0-, $\frac{1}{3}$- and $\frac{2}{3}$-mesh-sizes in all periodic directions. We use P_x with $x \in \{0, \frac{1}{3}, \frac{2}{3}\}$ to indicate that momentum vector $\mathbf{k}_p$ is sampled in $\mathcal{K}_x$ or the x-mesh. One useful rule is that for $\mathbf{k}_1 \in \mathcal{K}_x$ and $\mathbf{k}_2 \in \mathcal{K}_y$ with $x \neq y \in \{0, \frac{1}{3}, \frac{2}{3}\}$, $\mathbf{k}_1 + \mathbf{k}_2 \in \mathcal{K}_{2z}$ with $z = \{0, \frac{1}{3}, \frac{2}{3}\} \setminus \{x, y\}$. Note that when the subscript x is outside $[0, 1)$, $\mathcal{K}_x$ refers to $\mathcal{K}_{x-[x]}$ where $x - [x]$ is the fractional part of x.

3.1 Energy calculation

Consider the mesh sampling for the MP3 energy calculation with exact amplitudes in Eq. (2.2). Due to the two ERIs, $\langle IJ|AB\rangle$ and $\langle IJ|BA\rangle$, we need to sample both $\mathbf{k}_a$ and $\mathbf{k}_b$ on meshes different from $\mathbf{k}_i$. With three staggered meshes, there are two possible forms:

$$\frac{1}{N_{\mathbf{k}}^3}\sum_{I_xJ_xA_yB_z}\left(2\langle I_xJ_x|A_yB_z\rangle - \langle I_xJ_x|B_zA_y\rangle\right)t_{I_xJ_x}^{A_yB_z}(\infty) \tag{3.1}$$

$$\frac{1}{N_{\mathbf{k}}^3}\sum_{I_yJ_zA_xB_x}\left(2\langle I_yJ_z|A_xB_x\rangle - \langle I_yJ_z|B_xA_x\rangle\right)t_{I_yJ_z}^{A_xB_x}(\infty) \tag{3.2}$$

where $\{x, y, z\}$ can be any permutation of $\{0, \frac{1}{3}, \frac{2}{3}\}$ and $\sum_{I_xJ_xA_yB_z}$ means

$$\sum_{I_xJ_xA_yB_z} = \sum_{\mathbf{k}_i\in\mathcal{K}_x,\mathbf{k}_j\in\mathcal{K}_x,\mathbf{k}_a\in\mathcal{K}_y,}\sum_{ijab}$$

Note that the momentum conservation will uniquely determine the uniform mesh $\mathcal{K}_z$ for $\mathbf{k}_b$ when the meshes $\mathcal{K}_x, \mathcal{K}_x, \mathcal{K}_y$ for $\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a$ are provided.

3.2 Amplitude calculation

The staggered-mesh energy calculation above requires MP3 amplitudes sampled as $t_{I_xJ_x}^{A_yB_z}(\infty)$ or $t_{I_yJ_z}^{A_xB_x}(\infty)$. In the following discussion, we focus on a fixed set of (I, J, A, B) that are sampled on $\mathcal{K}_x, \mathcal{K}_x, \mathcal{K}_y$, and $\mathcal{K}_z$, respectively, and consider approximating $t_{I_xJ_x}^{A_yB_z}(\infty)$ using the staggered mesh method.

4h2p term in Eq. (2.3) Ignoring the constant fractional term $1/\epsilon_{IJAB}$, the 4h2p amplitude term converges in the TDL as

$$\frac{1}{N_{\mathbf{k}}}\sum_{KL}\langle KL|IJ\rangle\, s_{KL}^{AB} \longrightarrow \frac{1}{|\Omega^*|}\int_{\Omega^*}\mathrm{d}\mathbf{k}_k \sum_{kl}\langle KL|IJ\rangle\, s_{KL}^{AB}.$$

To approximate this integral by a trapezoidal rule while avoiding zero momentum transfer, we need to sample K on a mesh different from I and A. Since I and A are on x- and y- meshes respectively, K can be sampled on z-mesh and L is then sampled on y-mesh by momentum conservation. The staggered mesh method to approximate this term in the TDL can thus be defined as

$$\frac{1}{N_{\mathbf{k}}}\sum_{K_zL_y}\langle K_zL_y|I_xJ_x\rangle\, s_{K_zL_y}^{A_yB_z}.$$

2h4p term in Eq. (2.3) The 2h4p amplitude converges in the TDL as

$$\frac{1}{N_{\mathbf{k}}}\sum_{CD}\langle AB|CD\rangle\, s_{IJ}^{CD} \longrightarrow \frac{1}{|\Omega^*|}\int_{\Omega^*}\mathrm{d}\mathbf{k}_c \sum_{cd}\langle AB|CD\rangle\, s_{IJ}^{CD}.$$

To avoid zero momentum transfer, we need to sample C on a mesh different from those of A and I, and thus on z-mesh. By momentum conservation, D needs to be sampled on y-mesh. The staggered mesh calculation of this term is thus computed as

$$\frac{1}{N_{\mathbf{k}}}\sum_{C_zD_y}\langle A_yB_z|C_zD_y\rangle\, s_{I_xJ_x}^{C_zD_y}.$$

3h3p terms in Eq. (2.3) The summation of four 3h3p amplitude terms converges in the TDL as

$$\begin{aligned}&\frac{1}{N_{\mathbf{k}}}\sum_{KC}(2\langle AK|IC\rangle - \langle AK|CI\rangle)s_{KJ}^{CB} - \langle AK|IC\rangle\, s_{KJ}^{BC} - \langle AK|CJ\rangle\, s_{KI}^{BC}\\ &\quad\longrightarrow \frac{1}{|\Omega^*|}\int_{\Omega^*}\mathrm{d}\mathbf{k}_k\sum_{kc}(2\langle AK|IC\rangle - \langle AK|CI\rangle)s_{KJ}^{CB} - \langle AK|IC\rangle\, s_{KJ}^{BC} - \langle AK|CJ\rangle\, s_{KI}^{BC}.\end{aligned}$$

To avoid the zero momentum transfer, K and C need to be sampled on y-mesh and z-mesh, respectively. The staggered mesh method for this term can thus be defined as

$$\frac{1}{N_{\mathbf{k}}}\sum_{K_yC_z}(2\langle A_yK_y|I_xC_z\rangle - \langle A_yK_y|C_zI_x\rangle)s_{K_yJ_x}^{C_zB_z} - \langle A_yK_y|I_xC_z\rangle\, s_{K_yJ_x}^{B_zC_z} - \langle A_yK_y|C_zJ_x\rangle\, s_{K_yI_x}^{B_zC_z}.$$

For the symmetrized version of this term created by $\mathcal{P}$, K and C need to be sampled on z-mesh and y-mesh, respectively.

Gathering all these designs above, the staggered mesh method for MP3 amplitude is defined as

$$
\begin{aligned}
&t_{I_xJ_x}^{A_yB_z}(N_{\mathbf{k}}) \qquad (3.3)\\
&= s_{I_xJ_x}^{A_yB_z} + \frac{1}{\varepsilon_{I_xJ_xA_yB_z}}\Bigg[\frac{1}{N_{\mathbf{k}}}\sum_{K_zL_y}\langle K_zL_y|I_xJ_x\rangle\, s_{K_zL_y}^{A_yB_z} + \frac{1}{N_{\mathbf{k}}}\sum_{C_zD_y}\langle A_yB_z|C_zD_y\rangle\, s_{I_xJ_x}^{C_zD_y}\\
&+ \frac{1}{N_{\mathbf{k}}}\sum_{K_yC_z}(2\langle A_yK_y|I_xC_z\rangle - \langle A_yK_y|C_zI_x\rangle)s_{K_yJ_x}^{C_zB_z} - \langle A_yK_y|I_xC_z\rangle\, s_{K_yJ_x}^{B_zC_z} - \langle A_yK_y|C_zJ_x\rangle\, s_{K_yI_x}^{B_zC_z}\\
&+ \frac{1}{N_{\mathbf{k}}}\sum_{K_zC_y}(2\langle B_zK_z|J_xC_y\rangle - \langle B_zK_z|C_yJ_x\rangle)s_{K_zI_x}^{C_yA_y} - \langle B_zK_z|J_xC_y\rangle\, s_{K_zI_x}^{A_yC_y} - \langle B_zK_z|C_yI_x\rangle\, s_{K_zJ_x}^{A_yC_y}\Bigg].
\end{aligned}
$$

3.3 Madelung-like correction

In addition to the staggered mesh method, we still need a Madelung-like correction to remove $1/|\mathbf{q}|^2$ singular terms in the MP3 amplitude calculation because the quadrature error due to such singular terms not only originates from its omission at $\mathbf{q}=\mathbf{0}$ but also from the quadrature in all volume elements near $\mathbf{q}=\mathbf{0}$.

Consider the 4h2p term in Eq. (3.3) as an example:

$$
\frac{1}{N_{\mathbf{k}}}\sum_{\mathbf{k}_k\in\mathcal{K}_z}\sum_{kl}\langle K_zL_y|I_xJ_x\rangle\, s_{K_zL_y}^{A_yB_z} \longrightarrow \frac{1}{|\Omega^*|}\int_{\Omega^*}\mathrm{d}\mathbf{k}_k\sum_{kl}\langle KL|IJ\rangle\, s_{KL}^{AB}.
$$

The numerical quadrature over the leading singular term in this calculation can be reformulated as

$$
\frac{1}{N_{\mathbf{k}}}\sum_{\mathbf{q}\in\mathcal{K}_{x-z}}\frac{4\pi}{|\Omega|}\frac{1}{|\mathbf{q}|^2}s_{IJ}^{AB}, \qquad \mathbf{q}=\mathbf{k}_i-\mathbf{k}_k.
$$

Using singularity subtraction technique, we define the Madelung-like correction for this calculation as

$$
\frac{1}{N_{\mathbf{k}}}\sum_{\mathbf{k}_k\in\mathcal{K}_z}\sum_{kl}\langle K_zL_y|I_xJ_x\rangle\, s_{K_zL_y}^{A_yB_z} - s_{IJ}^{AB}\left(\frac{1}{N_{\mathbf{k}}}\sum_{\mathbf{q}\in\mathcal{K}_{x-z}} - \frac{1}{|\Omega^*|}\int_{\Omega^*}\mathrm{d}\mathbf{q}\right)\frac{4\pi}{|\Omega|}\sum_{\mathbf{G}\in\mathbb{L}^*}\frac{e^{-\sigma|\mathbf{q}+\mathbf{G}|^2}}{|\mathbf{q}+\mathbf{G}|^2}
$$

where $\sigma>0$ can be heuristically selected. Denote the above correction term as $-s_{IJ}^{AB}\xi_{x-z}$. Similarly, we define the Madelung-like corrections to the other 5 terms in Eq. (3.3) as

$$
\begin{aligned}
&\frac{1}{N_{\mathbf{k}}}\sum_{C_zD_y}\langle A_yB_z|C_zD_y\rangle\, s_{I_xJ_x}^{C_zD_y} \longrightarrow (\cdots) - s_{IJ}^{AB}\xi_{z-y},\\
&\frac{1}{N_{\mathbf{k}}}\sum_{K_yC_z}\langle A_yK_y|C_zI_x\rangle)s_{K_yJ_x}^{C_zB_z} \longrightarrow (\cdots) - s_{IJ}^{AB}\xi_{z-y},\\
&\frac{1}{N_{\mathbf{k}}}\sum_{K_yC_z}\langle A_yK_y|C_zJ_x\rangle\, s_{K_yI_x}^{B_zC_z} \longrightarrow (\cdots) - s_{IJ}^{AB}\xi_{z-y},\\
&\frac{1}{N_{\mathbf{k}}}\sum_{K_zC_y}\langle B_zK_z|C_yI_x\rangle\, s_{K_zJ_x}^{A_yC_y} \longrightarrow (\cdots) - s_{IJ}^{AB}\xi_{y-z},\\
&\frac{1}{N_{\mathbf{k}}}\sum_{K_zC_y}\langle B_zK_z|C_yJ_x\rangle\, s_{K_zI_x}^{C_yA_y} \longrightarrow (\cdots) - s_{IJ}^{AB}\xi_{y-z}.
\end{aligned}
$$

Applying all the Madelung-like corrections above to Eq. (3.3) and noting that $\mathcal{K}_{x-z} = \mathcal{K}_{z-y}$, we have the final staggered mesh method for amplitude calculation as

$$\widetilde{t}_{I_x J_x}^{A_y B_z}(N_{\mathbf{k}}) = t_{I_x J_x}^{A_y B_z}(N_{\mathbf{k}}) + \frac{2}{\varepsilon_{I_x J_x A_y B_z}} s_{IJ}^{AB} \xi_{y-z}. \tag{3.4}$$

Combining Eq. (3.4) and Eq. (3.1), the overall staggered mesh method for MP3 energy calculation can be defined as

$$E_{\text{MP3}}^{\text{stagger}}(N_{\mathbf{k}}) = \frac{1}{N_{\mathbf{k}}^3} \sum_{I_x J_x A_y B_z} \left(2 \langle I_x J_x | A_y B_z \rangle - \langle I_x J_x | B_z A_y \rangle\right) \widetilde{t}_{I_x J_x}^{A_y B_z}(N_{\mathbf{k}}).$$

A similar staggered mesh method can also be derived from the other energy calculation formula in Eq. (3.2).

References

[1] X. Xing, X. Li, and L. Lin. Unified analysis of finite-size error for periodic hartree-fock and second order m {\o} ller-plesset perturbation theory. *arXiv preprint arXiv:2108.00206*, 2021.